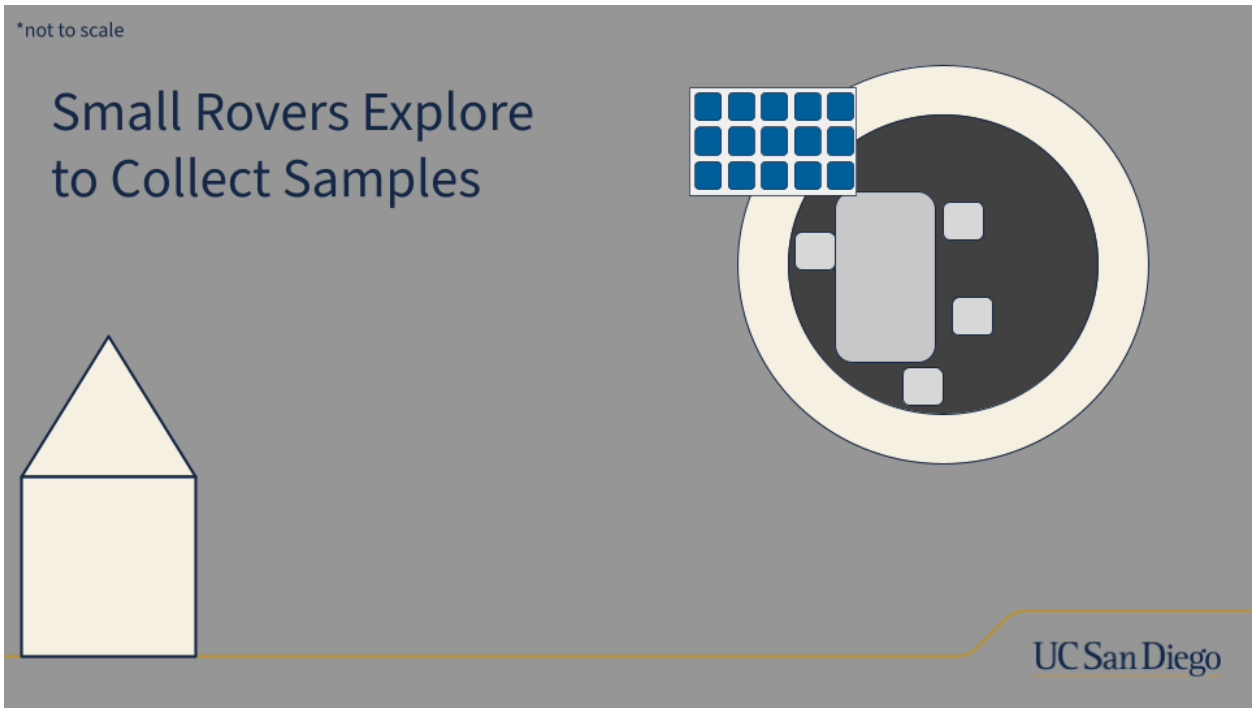
University of California, San Diego

Modular Autonomous Rovers for Collection of Samples (MARCS)

RASC-AL 2025, Theme: Lunar Sample Return Concept

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22, February 2026

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Theme: Lunar Sample Return Concept

Objectives & Technical Approach

- Provide researchers with largest volume of lunar samples whilst preserving traits
- Allow variety in types of lunar samples that can be obtained
- Rely on automation
- A network of rovers, each fitted to collect their "assigned" samples.

Image



Key Design Details & Innovations:

- Regolith 3D Printed containers reducing resupply mass and enabling long duration operations including QR codes containing data such as location coordinates
- AEGIS inspired autonomy which we scaled to a multi rover system
- Queen-Child rover system
- Polar optimized power system using solar towers and fuel cell backup which is aligned with Artemis strategies.
- Centralized 'Sample Library"

Schedule

- 2029: Early Development and Prototyping
- 2031: Integrated Rover testing on Earth
- 2033: Orbital Demonstrations
- 2034: Lunar Tech Demo (Small Scale)
- 2035: Full System Integration
- 2036: Lunar Surface Deployment

Cost

- Projected mission cost: $410-590 million
- Projected development cost: $780 million-$1.23 billion

1. Abstract

Our project encapsulates a system of rovers and a hub that acts as a central storage system preserving samples in a library-like manner to address sample collection, storage, and return. Samples in the hub are labeled in a clear manner, allowing for easy identification for the transport back to Earth. This transport process utilizes CLPS as a means to transfer samples from one location to another by leveraging private companies that work in tandem with NASA rather than requiring a large scale moon-to-earth sample return mission. Many of the methods for power management that are addressed in our paper easily transfer to the Mars surface, and are designed to align with NASA's moon to mars missions.

2. Project Introduction

Our rover system architecture is scalable and modular to support the vision of sustained lunar surface exploration and sample collection. The system is set to be in action during the early phases of Artemis Moon-to-Mars Campaign, in between Artemis IV through VIII that are planned to take place from 2030 to 2040. Our design is engineered for human-machine teaming while primarily operating with high level autonomy. The rovers support low-latency teleoperation and manual sample hand-off, allowing crew members to intervene for high-grading or complex tactical maneuvers if necessary.

The proposed architecture deploys a single primary unit (Queen) and four high mobility secondary units (Children), similar to the mother-daughter system. This 1:4 ratio is optimized for increased surface coverage while maintaining ample access to each rover type.

Our sample storage will be implemented via a 3D printed swappable container system that allows rovers to transfer their sample over to the sample library or hub in a quick and efficient manner. Each drop-off involves the sample storage container in the Queen being moved into the hub and replaced with a fresh and empty container identical in shape and size.

The rovers will also be equipped with computer vision models that allow real-time hazard detection and navigation on uncertain terrain, as well as key sample identification in live conditions.

For our power, we will be operating mainly in the polar regions as that seems to be the main areas of interest for the Artemis. This means we can strategically place solar towers in areas with peaks of eternal light. that use lithium-ion batteries for short-term energy storage. For areas of further expansion away from the poles, we will need to use fuel cells for long term energy storage. During the lunar day, we can split water to form liquid hydrogen and oxygen. Then once we need power later, we can ignite the fuels to get more power.

3. Timeline:

Year Range	***Phase***
2027- 2029	Early Development and Prototyping
2029-2031	Integrated Rover testing on Earth
2031-2033	Orbital Demonstrations
2033-2034	Lunar Tech Demo (Small Scale)
2034-2035	Full System Integration
2035-2036	Lunar Surface Deployment

2027-2029: Early Development and Prototyping
During this period we will develop the core rover architectures, including its mobility systems, thermal control and the sample collection mechanisms. We will also build the autonomy stack and have it tested in simulation focusing on computer vision, target selection and hazard detection. Early prototypes of our temperature controlled sample containers are produced and tested.
2029-2031: Integrated Rover Testing
From 2029-2031, the system transitions into integrated rover testing. A full scale Queen rover prototype will be assembled and tested. Field trials in lunar-like terrain such as the one at NASA's Ames Research Center will test its performance, navigation systems and the sample collection workflow. This phase is also where we integrate the short and long range communication hardware and the sample collection container designs will be advanced through its thermal vacuum testing.
2031-2033: Orbital Demonstrations
Here, we will conduct orbital and CLPS based demonstrations to ensure our subsystems work in relevant environments. A small scale autonomy and also a communications payload is flown on a CLPS mission to test operations including the store and onward data handling method that is mentioned in the communications section.. Thermal container performance and radiation shielding is also evaluated and the power system components such as the solar tower elements will be tested.
2033-2034: Lunar Tech Demo
Our system will perform a small scale lunar technology rehearsal. A small Queen rover and a single sample collection rover will be delivered for a limited surface test mission. This will help us validate sample pickup, our communication systems, the QR based sample tracking and mobility in shadowed regions. A successful mission here confirms readiness for a full scale integration.
2034-2035: Full System Integration

The Queen and four sample collection rovers are assembled and integrated as a completed system. This phase will include an immense amount of testing of the sample tracking workflow, the data analysis process, and the sample containers. The system will also be integrated with Artemis infrastructure like Gateway's communications and the sample library interface. Final testing and validation ensures that our system can operate reliably during a lunar surface mission.

2035-2036: Full Deployment

Finally, the complete system is deployed on Aetmeis VII or VIII. These missions provide the first sustained surface operations and cargo capability required for a multi rover sample return system. Once delivered the system becomes an operational asset during Artemis science campaigns. In all, this timeline proves a path to TRL-8 readiness by 2035 and TRL-9 validation during the Artmeis VII and VIII missions in 2036 and beyond.

4. Rover Roles

In order for the Queen rover and small rovers to be able to transverse over the terrain effectively, they will all have spring tires made from Nickel Titanium. Spring tires are resilient enough to withstand large impacts before deformation occurs. Additionally, they will all have a multilayer insulation coating on the outside made from aluminized Kapton and Beta cloth or more commonly known as Teflon coated fiberglass. This type of material was used extensively on Apollo and is used on modern lunar hardware as this coating will allow for the protection of both the Queen and collection rovers from the radiation experienced on the surface, providing thermal stability and durability across the Moon's harsh environment. Similar to past rovers, they will be made from past Titanium as it has a high stress-weight ratio and is stable across temperatures. Passive thermal control is implemented by either electrical resistance heaters or optional radioisotope heater units which ensure the electronics like the batteries or sample containers remain within operation temperature limits during lunar night. RHU's use of the natural decay of plutonium 238 to provide a continuous and steady heat source without requiring electrical power is a great update we can make for our rovers. This has been used on various missions such as Apollo, Voyager, Curiosity where maintaining the electronics was essential for surviving in deep cold environments.

The Queen rover's main purpose is to transfer the small rovers from location to location. As a result, its design will be centered around 1) providing a safe "room" for the small rovers to be transported without worry for hazards from the outside, and 2) having the power to transport these rovers over long distances. The Queen rover will have a "trunk" large enough to fit all the small rovers which will apparently be 3.0 m x 2.1m x 1.2 m, providing the necessary space to house all four sample collection rovers in a 2x2 configuration along with the structural supports, and insulation. A large lithium ion battery will be there as well in order to provide power for traveling from location to location. The rover will have a 5-6 kWh lithium ion battery, allowing the Queen to bring the small rovers up to 5-7 miles away from the main base where samples will be brought to later on.

Additionally, it will also have a dedicated antenna for continuous communication with the small rovers with additional details provided in section 8 of this proposal.

The small rovers' main purpose is to leave the Queen rover at arrival of the selected location of interest, explore the area for samples of interest, collect them, and then return back to the Queen rover to be transported back to the main base. The rovers will have a core drill bit in order to be able to extract cylinders of samples from the ground, and will then have a system similar to that of curiosity's samples processing in order to make sure the samples aren't contaminated. These rovers will then have a vacuum storage container made of regolith which will have a total volume of $0.1m^3$, allowing for up to 190kg of regolith to theoretically be stored . In order to identify potential samples, we'll have a camera which we selected as a 12 MP shutter camera as it provides sufficient resolution for identifying samples, assessing the terrain ahead and being able to detect hazards while remaining low power and light weight. Cameras such as this are widely used during NASA missions, including Ingenuity's 13MP RTE camera for capturing high resolution images. Also, we would include two 0.5 - 1 MP cameras to support navigation as these cameras would help track surface features, Additionally, the small rover will be equipped with a 2.4 Ghz low power antenna in order to send and receive signals from the Queen rover. This enables continuous huh two way communication between the rovers which allows for navigation updates, reporting on rover health and sample collection data. This link also supports 1-2 Mbps, which will allow our rover to transmit images. In order to process data, the Queen sends out information relating to identifying samples, and the small rover will have a small chip to process said information. The small rovers will be equipped with a radiation tolerant ARM Cortex-M7 microcontroller which will be paired with a lightweight FPGA vision module. The Cortex M7 will handle high level logic such as interpreting the Queen's sample priorities and updating its navigation plans, as well as managing the communication protocols. On the other hand the FPGA will perform fast low power image processing such as detecting the edges of a sample, analyzing the texture and more specific sample parameters using that 12MP camera. This combination allows the rover to autonomously evaluate its surroundings, identify the potential sample, and begin a routine that avoids hazards while maintaining low power consumption and high reliability in a lunar environment. In all, this rover architecture is designed to extend the technologies that NASA has already validated on previous lunar missions, as NASA and CLPS providers have used this radiant tolerant ARM microcontrollers for low power control on missions before.which will mean that failure during a mission is very low.

5. Sample Identification

Currently, collecting samples requires a large amount of manual effort as sample collection needs to be planned out greatly in advance. But with the amount of rovers in our network, we can't rely on such laborious methods before every single outing. However, this weakness is also our strength. If we have enough rovers scanning an area of interest, we can continue to search for appropriate target

samples over a wider search area, minimizing the amount of planning needed to approach a single, precise location.

To not just offload the labor onto controlling each of the rovers, each rover will be running a modified version of AEGIS, which has already been used to select target samples on Curiosity. Currently AEGIS mainly segments and identifies features within images, then computes whether it should fire a laser based on a fixed heuristic. We want to continue using traditional computer vision algorithms to extract features, but we also want to run this data against a stakeholder-defined metric, rather than an algorithm defined at launch. This flexible weighting system means that each individual rover can be very specific about which samples it is searching for and can even be swapped from excursion to excursion.

Once the rover is on the ground, it will search its designated area, periodically collecting data from the lunar surface beneath it and extracting those aforementioned features. After running the data through its weighting system, if the rover determines that it finds a suitable sample candidate, it will send an image back to mission control for human review. This means the rover is doing the laborious work autonomously, while maintaining granular control over whether we should actually collect this sample, whether we need more images for more information, or reevaluate the weighting algorithm.

Once mission control signs off, then the rover will begin its sample collection.

We recognize these rovers will have very limited compute power, so we have intentionally shied away from using heavier convolutional networks. Additionally, the CNNs work as a black box. We can't scrutinize why the AI had concluded this feature exists. But if we ensure that we design our traditional computer vision algorithms, we can understand the data extracted and check the rover's work if needed, rather than blindly trusting an AI system.

Sample organization

Sample storage will be held in a central hub which acts as a library. Samples will be tagged with a QR code that identifies and stores information about the samples such as the location it was collected from, and on or underground. This allows for clear organization for retrieval purposes. Astronauts on the moon and scientists on the earth can use the marked samples for ease of access. The storage of these samples will leverage new research findings from the lunar freezer system currently in development by University of Alabama at Birmingham. This sample organization allows the current and future crew and scientists to easily extract data about sample collection, and contents, and offers a clean way to target samples to be sent back to Earth for future study.

6. Power

With our hub and spoke model of sample collection, we will have a lot of scattered nodes of areas of interest. With our transportation grid looking like a sparse map, it's in our best interest to minimize the number of stationary structures, as they won't get reused often.

Our main strategy for power is to strategically place solar towers near areas of interest. The idea is that the rovers are able to resupply power here and also drop off samples. So far, we are unable to manufacture solar panels on the moon, so our initial solar panels must be flown from the Earth. We are hoping to have these solar panels to be perovskite based, or perhaps a perovskite-silicon blend. This has the advantage of being not only lighter, but also has greater efficiency and resilience. While these perovskite solar panels are currently only at a TRL of 4 as of writing, there's confidence that these perovskite-silicon blends can be space-ready with a TRL of 8 or 9 by launch by 2035.

Additionally, if we only place our solar towers near peaks of eternal light, our towers can be equipped mainly with short-term energy storage, such as lithium-ion batteries. Rovers could then come over and just charge up. If we want longer-term energy storage, we should use the abundant ice in the south pole. We can use solar power to split hydrogen and oxygen. A fuel cell can later recombine them for power.

While solar is renewable which is important for these long term missions, significant power developments in powerbeaming would help increase the robustness of our system. By putting a few satellites in orbit that ride the terminator line, they can complement or even overtake these stationary power stations, as they are able to provide power to large areas of the moon simply by swiveling a few degrees. This negates the need for solar panels on the ground that require maintenance and significant transport costs.

7. Communication

In our mission, we are focusing on a multi-rover lunar sample return architecture. We rely on a transport rover which we are calling the "Queen" who will deploy multiple smaller collection rovers to autonomously retrieve samples of lunar material. Therefore, our communications design must support coordinated swarm operations, long range traverses and continuous sample - health monitoring. Thus, we are approaching this by using a hierarchical relay structure, placing long-range communication hardware on the Queen rover while keeping our collection rovers simple and lightweight to improve efficiency.

Each collection rover will communicate with our Queen rover using a 2.4 GHz link which will provide a short range, low power radio connection for local data exchange between the Queen and collection rovers. Through this, the small rovers will be able to send health updates, navigation data, sample collection data, temperature or shock reading and much more. What this does is that it allows the Queen to act as a central controller, keeping a record of every sample and managing tasks. This allows power use to be low which makes these rovers efficient and focused on the task at hand which is collecting these samples.

Next, when the Queen reaches the lunar sample hub, it switches to a fast surface connection that will allow it to transfer large amounts of data, including sample health logs, images, and data about what we discover on the moon. The hub serves as the mission's central collection for both samples and sample data, which allows long term storage in a stable environment. While the small rovers physically deliver the samples, the Queen will be able to deliver the complete digital record of each sample to the hub through the surface link. However, we can also account for long range communication, where the Queen rover employs S-band or X-band radios to communicate with Gateway when it has line of sight of it. This allows Gateway to relay mission data back to Earth through the Deep Space Network which also allows mission control to send updated instructions back to the Queen. This approach follows current NASA practices and helps by avoiding communication directly from the lunar surface to Earth which would call for high power transmissions and software.

Now because the rovers may operate in terrain that may block the “line of sight”, the system will use a store and forward strategy. During these blackout periods, such as when the Queen is behind terrain or operating on the lunar far side, both the transport and collection rovers will log all telemetry and the sample’s data to onboard storage and once connectivity is restored, the stored data is transmitted in bulk ensuring a complete record of rover activity and sample history.

8. Store +Transfer the samples

Each of the small rovers will collect the samples by drilling them out with a core drill bit. This will allow us to preserve the structure of the sample material so that it could be observed later. Similar methods have been used on curiosity rovers in the past to collect samples, so this feature is already ready to be utilized. The rover would then be able to pick up the material and then place it into a vacuum insulated chamber, where it would be able to keep it at a constant temperature for extended periods.

We will have two main types of containers:, one for cryogenic storage, one for temperature controlled samples. The cryogenic storage container will be only concerned with maintaining the cold temperature inside, so it can just be designed as a vacuum insulated container meant for transport no longer than a day. Meanwhile, the temperature controlled samples require a bit more work. We will need a resistive heating element inside the container that can be turned on in order to add back heat in case any gets lost. This means each of these temperature controlled containers will require its own power supply and heat sensor to ensure the sample maintains at its individual desired temperature. The temperature controlled containers will also be radiation shielded by polyethylene lining, so that our mineral samples will remain undisturbed and that our biological samples won’t be harmed.

The small rovers would have multiple chambers so that it could collect a variety of samples without contaminating the other samples. After collecting its assigned samples, each of these rovers will then return to regroup with the Queen rover. After all of the collection rovers return, they will then be transported back to the main base where the Artemis missions will be landing. At the main base, there will be a “library”, where the smaller rovers will be able to leave the samples in their containers to be

stored for an extended period for researchers to be able to study. At the base, a new replacement for the container will be given in order to prevent cross contamination and allow for easier movement for samples. After receiving new containers, the small rovers would return to the Queen rover and head to the site of interest.

The larger storage facility will be made alongside the lunar surface habitat. Parts could be imported from earth alongside with the Artemis VIII mission. Additionally, a 3D printer capable of printing with regolith should also be brought from earth. This would enable the production of new vacuum storage containers. This would solve two problems: firstly, one wouldn't have to worry about cross contamination of samples when reusing such containers, as a new container could be used after every mission, and secondly, a library of samples to be kept on the moon without worry of cross contamination, and could be saved for later transportation of said samples to earth if deemed necessary.

9. System Cost:

Rover Development Cost:

The cost for our Queen and sample collection rovers is estimated using NASA rover analogs such as VIPER that although did not launch still provides relevant data and the rovers used for the Mars exploration missions. The VIPER rover was canceled due to cost overruns and pressure in the scheduled time frame but its published cost provides a reasonable approximation for our rover . Our queen rover is significantly simpler as well as it is used mainly for a transport vehicle rather than having complex systems like the complex drilling system or complex science payloads that VIPER would have, allowing us to estimate down to $250 - $300 million for construction. Each sample collection rover can be referenced by using the Mars rovers Spirit and Opportunity. Now the mars rovers were approximately $400 million each however that is including all their advanced science payloads and long range autonomy. Because our rovers are closer in regards to mass but far simpler in instrumentation, a realistic estimate is $40 to $60 million per rover which is consistent with CLPS robotic mobility systems. With the four sample collection rovers, the total development cost is roughly $170 to $240 million. Together the Queen rover and four sample collection rovers yield a development cost of $400-$590 million.

Sample Container Cost:

The sample container costs are taken in consideration from NASA's cryogenic integrated systems program that occurred during the Apollo missions. The sample containers used during the Apollo missions were approximately $150k in today's dollars, however our sample containers are much more advanced and rugged than the Apollo container's hardware. Thus we can estimate that per cryogenic unit, the cost would be about $200-$250k. Our containers, however, are more advanced in which they are temperature controlled and require heating elements, radiation shielding, and sensors

which would place them closer to the $250 to $400k range. With the use of about 40-60 containers across our mission, the total cost for sample handling hardware is about $10 to $17 million.

Communication Hardware Cost:

Communication subsystems are based on prior costs for missions such as Mars Reconnaissance Orbiter as well as VIPER, which fall in the 15 to 30 million range. This is based on NASA's publicized hardware budgets. With this in mind, an estimate for the Queen's long range communication systems can be about 15 to 25 million. For our sample collection rovers, small rover radio packages used in CLPS payloads typically cost $500k to $1.5 million per unit. Our sample collection rovers require short 2.4 GHz radios with systems like basic health monitoring sensors and so we can estimate about $1 to $2 million per rover which results in a total communications cost of $20-$31 million dollars.

Power Infrastructure Cost:

Our costs here are taken from NASA's solar array and power system programs. CLPS delivered solar arrays typically cost 10 to 25 million per deployment. However, because our solar towers will use perovskite- silicon panels and require structure hardware, a good estimate is 20 to 35 million per tower. Most NASA power programs cost roughly 8 to 12 million to develop, but scaling this to a lunar surface ready system yields an estimate of 10 to 15 million. In total, the power infrastructure is expected to cost anywhere from 70 to 160 million.

Launch and Lunar Delivery Cost:

The launch and delivery costs are based on NASA's Commercial Lunar Payload Services pricing and Artemis allocations. A recent CLPS award to Astrobotic shows a $79.5 million amount for sending 14 payloads to the moon in proportion for lunar missions which supported an estimate of $80 to $120 million for delivering our rovers. Also,for the Artemis VII/VII cargo, we use NASA's internal cost per kilogram, estimates for SLS Block 1b, which place cargo allocations in the $150 million range. Thus, we estimate a total delivery cost of about $250 to $430 million.

Total Cost:

Combining our rover development, the sample container development, our communication hardware, power infrastructure and lunar delivery costs we estimate a total mission cost of $780 million to $1.23 billion. This range is consistent with NASA robotic missions and reflects our complex multi rover and lunar sample return system. This cost structure aligns with past missions and provided a realistic estimate for our mission.

10. How concept addresses different needs of science stakeholders

Our main stakeholders would be space agencies such as NASA, research institutions and universities, and private companies looking to develop other lunar technologies.

Our concept addresses the needs of these stakeholders as we have set up an autonomous system with high sample throughput. Thanks to each excursion having four separate rovers, we can essentially divvy up sample selection spots through our stakeholders.

The actual interfacing with these organizations can be handled by NASA, but once these partners have submitted their sample requests, they can be batched up based on where they want to collect their samples from. All that then needs to be done on NASA's side would be to load up the scouting rovers with the appropriate sample containers, take them to the designated site via the queen rover and then collect all the samples requested by stakeholders. So even if stakeholders want specifically different samples, we can still optimize our collection missions by ensuring we don't have to return to a nearby area as often.

Because each set of queen rovers is independent of each other, we can have multiple autonomous missions occurring in parallel, allowing more stakeholder requests to be fulfilled.

11. Extensibility to Mars

Similar types of rovers can be deployed for future exploration missions on Mars. These rovers, similar to how they would be brought alongside the Artemis missions while they are transporting large objects, could be brought whenever a firm understanding regarding colonization of objects in space is developed from the Artemis missions. This would likely be in the far future after the Artemis missions, so this would likely occur at least by 2040 at a minimum.

12. Appendices

Works Cited

"Moon Missions." n.d. NASA Science. Accessed February 23, 2026.

https://science.nasa.gov/moon/missions/.

NASA. 2023. Power Beaming From Lunar Orbit to Small Lunar Science Assets.

https://ntrs.nasa.gov/api/citations/20230018523/downloads/Power%20Beaming%20From%20Lunar%20Orbit%20Final%20version.pdf.

NASA. 2024. "SLS (Space Launch System) Block 1B."

https://www.nasa.gov/reference/sls-space-launch-system-block-1b/.

NASA. 2026. "Polyethylene Production from In-Situ Resources in Microchannel Reactors." NASA TechPort. https://techport.nasa.gov/projects/90012.

NASA. 2026. “Regenerative Fuel Cell (RFC).” NASA TechPort. https://techport.nasa.gov/projects/116307.

Smith, Nancy. 2017. “Reinventing the Wheel.” NASA.gov. https://www3.nasa.gov/specials/wheels/.

NASA. n.d. “Communictaing with Curriosity.” NASA Science. https://science.nasa.gov/resource/communicating-with-curiosity-2/.

NASA. 2026. “Artemis.” NASA Humans in Space. https://www.nasa.gov/humans-in-space/artemis/.

NASA 2023. “The Lunar Lab and Regolith Testbeds.” NASA Ames Research Center https://www.nasa.gov/centers-and-facilities/ames/the-lunar-lab-and-regolith-testbeds/.

NASA 2025. “ Small Spacecraft Avionics”. State of the Art https://www.nasa.gov/smallsat-institute/sst-soa/small-spacecraft-avionics/